M378K: April 15th, 2026.

Hypothesis Testing.

Proof by Contradiction.

K ... the claim we're trying to PROVE to be true

Q: What if K were not true?

Assume $\boxed{\text{not } K}$

fact Q ..... fact(not Q)

These cannot coexist!

We say that we reached a contradiction!

$\Rightarrow\Leftarrow$

Our assumption of not K was wrong!

Hypothesis Testing.

Claim about parameters we're trying to SUBSTANTIATE.

μ ... the population mean parameter (say, the mean cholesterol level w/ pills)

μ_0 ... the null population mean (say, an existing benchmark) } A NUMBER

$\boxed{\mu < \mu_0}$ Alternative Hypothesis.

Assume $\boxed{\mu = \mu_0}$ Null Hypothesis.

⋮

collect data
statistical analysis

Figure out the $\boxed{\text{probability}}$ of seeing the data that we saw (or something more extreme) if $\mu = \mu_0$.

p-value

If the p-value is "small", we have evidence against $\mu = \mu_0$.

The smaller the probability, the STRONGER THE EVIDENCE.

The Normal Case w/ σ Known.

The Population Model: $Y \sim N(\boxed{\text{mean} = \mu}, \text{sd} = \sigma)$

unknown & of interest

Hypothesis Testing Procedure:

First: Set the Hypotheses

2nd Null Hypothesis: $\boxed{H_0: \mu = \mu_0}$

1st Alternative Hypothesis:

$$H_a: \begin{cases} \mu < \mu_0 & \text{(lower or left-sided)} \\ \mu \neq \mu_0 & \text{(two-sided)} \\ \mu > \mu_0 & \text{(upper or right-sided)} \end{cases}$$

Second: Figure out the appropriate TEST STATISTIC (TS).

Natural choice: $\bar{Y} \sim \text{Normal}\left(\text{mean} = \mu, \text{sd} = \frac{\sigma}{\sqrt{n}}\right)$

Under the null hypothesis, i.e., for $\mu = \mu_0$,

$$Z = \boxed{\frac{\bar{Y} - \mu_0}{\frac{\sigma}{\sqrt{n}}} \sim N(0,1)}$$

Third: Consider the observed value of the TS.
In this case, it's $\bar{y}$, i.e., the observed sample average.

Q: What is the probability of observing $\bar{y}$ or something $\boxed{\text{more extreme}}$ under the null?

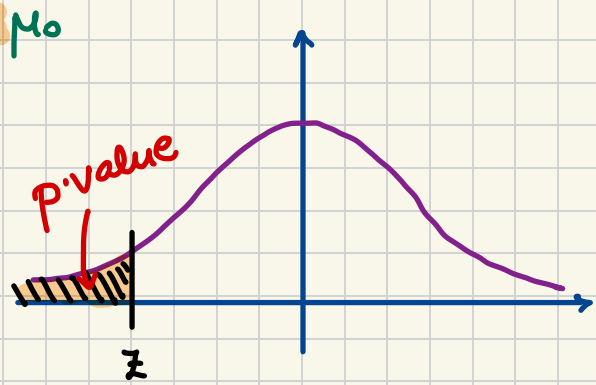
Exact interpretation depends on the structure of the alternative hypothesis. ✓

Regardless:

$$\boxed{z = \frac{\bar{y} - \mu_0}{\frac{\sigma}{\sqrt{n}}}}$$

Left-Sided Alternative.

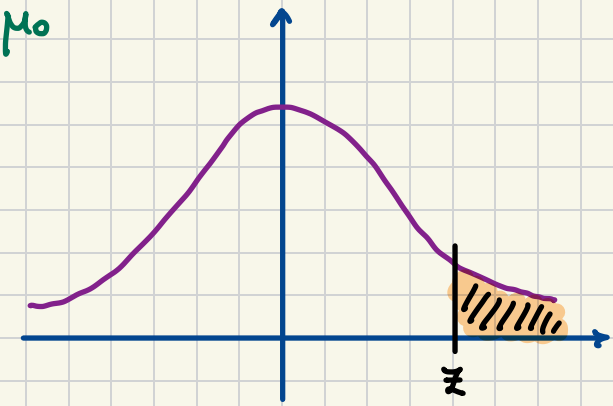
$H_a: \mu < \mu_0$



$$\text{p-value} = \Phi(z) = \text{pnorm}(z)$$

Right-Sided Alternative.

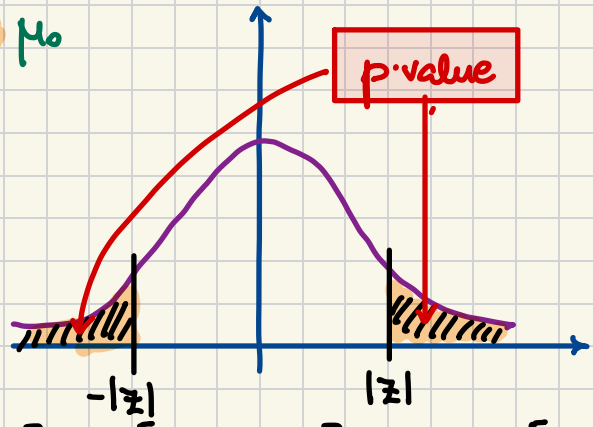
$H_a: \mu > \mu_0$



$$\text{p-value} = \mathbb{P}[Z \geq z] = 1 - \Phi(z) = \Phi(-z) = \text{pnorm}(-z)$$

Two-Sided Alternative.

$H_a: \mu \neq \mu_0$



$$\begin{aligned}
\mathbb{P}[Z \geq |z|] + \mathbb{P}[Z \leq -|z|] &= 2 \cdot \mathbb{P}[Z \geq |z|] \\
&= 2 \cdot \mathbb{P}[Z \leq -|z|] \\
&= 2 \cdot \Phi(-|z|) \\
&= 2 * \text{pnorm}(-\text{abs}(z))
\end{aligned}$$